

SOUTHERN ILLINOIS, IL, USA

WMO: 724336

Lat: 37.780N

Lon: 89.250W

Elev: 124

StdP: 99.85

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 93810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-9.8	-18.1	0.8	-11.3	-15.7	1.0	-8.8	12.3	10.3	11.3	9.5	3.1	330	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.2	24.9	33.0	24.7	32.1	24.4	26.7	32.0	25.9	31.2	25.3	30.3	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.6	29.9	24.2	19.4	28.9	23.7	18.8	28.3	84.0	31.6	80.6	31.5	77.7	30.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.6	8.9	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.9	36.4	2.9	1.8	-19.0	37.7	-20.7	38.7	-22.3	39.7	-24.4	41.0
			-17.1	27.9	3.0	0.6	-19.2	28.4	-21.0	28.7	-22.7	29.1	-24.8	29.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	1.0	3.1	8.1	14.2	19.5	24.1	25.6	24.7	20.7	14.4	8.0	2.8
	DBStd	9.94	6.11	6.04	5.86	4.87	4.23	3.10	2.77	2.72	3.83	4.99	5.35	5.78
	HDD10.0	962	287	204	106	17	1	0	0	0	0	14	99	233
	HDD18.3	2418	538	427	318	139	39	2	0	1	21	142	310	481
	CDD10.0	2391	7	11	49	143	294	423	484	457	322	152	40	10
	CDD18.3	807	0	0	2	15	75	175	226	199	92	21	1	0
	CDH23.3	7863	0	0	13	114	635	1756	2332	1898	901	206	8	0
	CDH26.7	3123	0	0	0	12	176	719	1022	808	334	52	0	0
Wind	WSAvg	3.2	3.9	4.0	4.0	4.1	3.3	2.8	2.3	2.1	2.1	2.8	3.5	3.6
Precipitation	PrecAvg	1164	79	71	106	127	132	109	97	80	83	81	108	89
	PrecMax	1677	167	155	291	380	252	245	182	192	243	204	217	207
	PrecMin	781	16	16	46	43	32	33	37	14	16	14	10	18
	PrecStd	209	37	34	54	70	60	55	45	37	47	49	59	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	21.2	25.2	28.2	32.1	34.8	37.4	36.5	34.1	30.7	24.5	19.5
		MCWB	16.1	16.5	17.6	19.1	23.1	24.3	25.0	25.0	22.7	22.2	17.4	16.7
	2%	DB	16.7	18.3	22.7	26.4	30.2	33.4	34.2	34.0	32.2	27.9	21.6	17.1
		MCWB	14.6	14.3	16.6	18.6	22.4	24.4	25.6	25.0	23.3	20.2	16.1	14.6
	5%	DB	13.6	15.9	20.4	24.4	28.7	32.3	32.9	32.5	30.5	25.7	19.1	14.3
		MCWB	11.0	12.4	15.1	17.7	21.5	23.9	25.5	24.6	22.7	18.7	14.8	11.8
	10%	DB	10.8	12.9	17.9	22.7	27.2	31.0	31.9	31.1	28.4	23.2	17.4	11.7
		MCWB	8.0	9.2	13.5	16.7	20.7	23.4	25.0	24.0	21.4	17.5	13.5	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.6	19.1	21.4	24.6	26.8	27.9	27.5	25.5	23.4	19.4	17.5
		MCDB	18.0	20.1	23.5	26.3	29.6	32.0	33.0	32.6	30.6	28.7	22.7	19.5
	2%	WB	14.9	15.3	17.6	20.0	23.6	25.7	27.0	26.3	24.5	21.9	17.4	15.0
		MCDB	16.8	17.5	21.8	24.5	28.6	31.4	32.2	31.9	29.7	26.1	19.6	16.8
	5%	WB	11.7	12.9	15.9	18.9	22.6	25.0	26.2	25.6	23.7	20.4	15.7	12.4
		MCDB	13.0	15.2	19.5	23.1	27.2	30.5	31.5	30.9	28.4	23.8	18.9	14.0
	10%	WB	7.9	9.6	14.2	17.7	21.6	24.2	25.5	24.8	22.8	18.7	13.7	9.2
		MCDB	10.1	12.3	17.3	21.3	25.8	29.3	30.4	29.7	26.9	22.2	16.9	11.3
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	10.7	11.6	11.1	11.1	10.6	11.5	13.1	12.8	11.2	8.9
		MCDBR	11.8	13.5	13.1	13.2	12.5	12.3	11.6	13.2	14.8	14.8	13.7	11.3
		MCWBR	9.9	10.3	8.5	7.3	6.0	5.0	4.8	5.6	6.6	7.8	9.2	9.3
	5% WB	MCDBR	10.4	12.3	11.6	11.0	10.8	10.7	10.4	11.4	11.9	12.3	11.5	10.0
		MCWBR	9.8	10.4	8.2	7.0	5.8	5.1	5.0	5.5	6.0	7.2	9.1	9.3
Clear-Sky Solar Irradiance	taub	0.314	0.325	0.363	0.395	0.429	0.438	0.473	0.468	0.408	0.357	0.320	0.306	
	taud	2.466	2.418	2.372	2.281	2.242	2.232	2.156	2.162	2.320	2.454	2.499	2.493	
	Ebn at Noon	871	905	898	884	856	845	812	807	839	857	857	855	
	Edh at Noon	82	97	112	131	138	140	150	146	117	93	79	75	
All-Sky Solar Radiation	RadAvg	2.10	2.77	3.79	4.99	5.62	6.44	6.21	5.67	4.73	3.52	2.38	1.77	
	RadStd	0.21	0.33	0.35	0.32	0.48	0.40	0.40	0.31	0.29	0.37	0.26	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77

Nomenclature: See separate page